Advanced Signal Filtering for Mean Reversion trading

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Abstract This paper investigates mean reversion trading using advanced neural signal filters. Mean reversion is the assumption that an asset's price will converge to a "fair value", like a simple moving average, over time. The spot asset price, influenced by high-frequency noise, may temporarily be above or below the fair value. Assuming the spot price will revert to the fair price in the long-term, this presents buy or sell opportunities for the reversion trading model.

We develop a regime-aware mean-reversion trading framework in which adaptive signal filters determine the latent fair price. We introduce an original optimisation objective; local average filtering objective (LAFO) interpolates between pointwise fitting and global averaging, yielding a controllable low-pass filter. We show how appropriate penalty terms restore identifiability, encode structural regime assumptions, and stabilise the extracted signal. The resulting spot–filter spread and its mean reversion properties can be directly linked to trading performance measures. We further show how modern neural network filters, including recurrent, convolutional, and state-space neural architectures, approximate solutions within this framework.

1 Introduction

Price data show trend-following or mean-reversion characteristics. Trend-following, or momentum, is based on the assumption that price trends are persistent - rising assets tend to continue rising, while those that are falling tend to keep declining. Mean reversion, in contrast, assumes that asset prices revert toward their long-term average over time. Empirical evidence from Bondt and Thaler (1985), Jegadeesh and Titman (1993), and Balvers and Wu (2006) shows that interday asset returns often exhibit momentum over short to medium horizons (three to twelve months) and mean reversion over longer horizons (three to five years). At higher frequencies, however, works such as Hasbrouck (2007) and Bouchaud et al. (2018) show this relationship tends to invert: microstructure effects such as bid–ask bounce, inventory control, and order-flow autocorrelation induce short-horizon mean reversion, while directional order flow can generate intraday momentum over slightly longer horizons. Therefore, algorithmic trading strategies can be broadly categorized into momentum and mean-reversion approaches, which apply statistical arbitrage to exploit these opposing market characteristics over different time horizons.

The mean reversion trading model exploits short-term price deviations around a "fair value", selling when the price is above it and buying when below. The interpretation of "fair value" depends on the assumed degree of market efficiency. Under weak-form efficiency, prices fully reflect past price information; under semi-strong efficiency, they additionally incorporate all publicly available information; and under strong-form efficiency, even private information is instantaneously reflected. While the efficient market hypothesis Fama (1970) provides a useful benchmark, both theory and evidence indicate that efficiency holds only approximately and in equilibrium. In this sense, prices reflect available information but contain economically small, noisy, and transient deviations that may be statistically detectable but are constrained by risk, transaction costs, and estimation error. Moreover, equilibrium efficiency implies that prices are driven by rational behaviour. New information acquisition, trading frictions, and capital constraints may move the price and not instantaneously reflect an asset's fundamental value. Grossman and Stiglitz (1980) show that perfect informational efficiency is impossible when information acquisition is costly, implying that observed prices remain noisy and incomplete signals of underlying value. Empirically, long-horizon mean reversion Poterba and Summers (1988) and slow capital adjustment with persistent order flow Gabaix and Koijen (2023) generate predictable price dynamics even under rational agents, while Malkiel (2003) argues that efficiency should be interpreted probabilistically rather than literally. Taken together, these results support weak-form efficiency at short horizons and approximate semi-strong efficiency in the long run, while rejecting strong-form efficiency.

Different from valuation approaches that anchor fair value to external economic or accounting variables, such as Ohlson (1995) accounting-based fundamental analyses, Fama and French (2015) and Hou et al. (2015) asset pricing, and Baker and Wurgler (2007) sentiment-based models, this work treats fair value as an internally defined latent price process inferred directly from market data. This is consistent with weak-form efficiency and equilibrium models in which prices are noisy signals of underlying value. On single, pairs, or portfolios of assets, we may decompose the price series (multiple assets are linearly combined) into a persistent fair-value component and a tradable mean-reverting residual, similar to Hasbrouck (2002). We follow modelling assumptions such as Beveridge and Nelson (1981) ARIMA and Kalman filtering decomposition, Engle and Granger (1987) and Gatev et al. (2006) cointegration/error-correction and pairs trading, and Hasbrouck (2002) state space price estimation using microstructure vector autoregression. Leung and Li (2015) and Lipton and Prado (2020) derive optimal threshold rules (entry/exit, profit/stop), treating data as Uhlenbeck and Ornstein (1930) processes. The fair price is a filtered trend signal, conceptually similar to moving averages but is more responsive, forward-looking, and adaptive to regime changes affecting intercept, trend, or autoregression coefficients under models such as Hamilton (1989) and Bai and Perron (1998, 2003).

We firstly introduce the local average filtering objective (LAFO). We show that its solution behaves as low-pass filters of selective frequency. The low-pass kernel weighting is dependent on the

LAFO window size K, which performs an interpolation between global averaging and pointwise mean squared error residual fit. We also show that a suitable penalty term can: 1. ensure smooth filtering output such that rebalancing and trading costs only occur when strongly supported by data, 2. recover uniqueness for LAFO's solution, 3. accommodate structural breaks and nonlinearity. Works on trend filtering such as Tikhonov and Arsenin (1977), Hodrick and Prescott (1997), Mammen and van de Geer (1997), Eilers (2003), S.-J. Kim et al. (2009), and Tibshirani (2014) impose structure on model output by varying the penalty term. Tibshirani (2014) reveals that the penalty leads to an output structure of piecewise polynomials of degree k with data-adaptive knots. The cases $k = 0$ (piecewise constant) and $k = 1$ (piecewise linear) are especially relevant for financial series under level-stationary or trend-stationary assumptions.

When selecting neural filtering models, we are presented with models of broadly two categories: those with and without state-space representations. Models without state space representations such as LSTMs and Transformers can learn signal extraction mechanisms directly from data. For instance, Horvath et al. (2025) show the efficiency of Transformers in non-linear and non-Markovian filtering. We compare model structures of Wavenet in Oord et al. (2016), Dual-Path RNN from Luo et al. (2020), and Informer by H. Zhou et al. (2021).

Meanwhile, Bayesian state space estimation frameworks such as Kalman (1960) and Gordon et al. (1993) are naturally methods of inferring latent variables, or hidden states such as the fair asset value, from noisy observations. The inference procedure recovers the unobserved latent variable using a combination of state transition equations (describing the latent dynamics) and observation/emission equations (linking latent states to observed data). Harvey (1990) and Hyndman et al. (2008) show that highly effective models, exponential smoothing and ARIMA, are naturally represented in and inferred through the state space framework. Hasbrouck (2002) filters for the fair price through signal noise decomposition (assuming the signal follows a random walk) with the state space inference structure. Works of Carlin et al. (1992), Geweke (1993), and S. Kim et al. (1998) demonstrate that the Bayesian estimation framework can cleanly and naturally encode stylised facts such as stochastic volatility, fat tails, and nonlinearity. In addition, discrete state space models such as the regime switching model by Hamilton (1989) and Switching Kalman filters by Murphy (1998) offer regime-conditional linear dynamics through latent Markov processes. Lastly, standard trend estimators such as Tikhonov and Arsenin (1977), Hodrick and Prescott (1997), and S.-J. Kim et al. (2009) are computed on the full sample. The standard trend estimators include future information, but when cast in state-space form they can be estimated recursively.

S. Gu et al. (2020) and Makridakis et al. (2020) show that expressive, nonlinear, data-driven machine learning solutions outperform purely statistical state space models, whose restrictive functional forms struggle to effectively capture market regime shifts, volatility clustering and other stylised facts in Cont (2001) and Ratliff-Crain et al. (2024). Krishnan et al. (2015) extends the

principles of Bayesian state-space estimation to neural-parameterised transition and emission models. Chung et al. (2015), Fraccaro et al. (2016), Kingma and Welling (2022), and Roque et al. (2025) explore alternative ways of conditioning the prior using RNNs, hierarchical structures, or Transformer-based models, while retaining the same underlying DKF-style latent state formulation. Rangapuram et al. (2018) show a scalable probabilistic state space forecasting method leveraging the data efficiency of state space models on both small and large regimes. Furthermore, recent innovative deterministic state space modelling architectures such as the State Space Sequence Model (S4) introduced by A. Gu et al. (2022) have demonstrated the potential to outperform Transformers in modelling long-range temporal dependencies.

We can therefore construct a mean reversion trading framework that acts on the spread between the spot and filtered price. The strategy can be formulated as a minimal and extensible optimal control problem. It is a simplified form of the optimal entry/take-profit structures studied in Leung and Li (2015) and Lipton and Prado (2020). This formulation links the statistical properties of the filtered signal, such as stationarity, smoothness, and rate of reversion, to trading performances such as Sharpe ratios, turnover, and excess returns. In line with optimisation approaches like Moody and Saffell (2001), Garleanu and Pedersen (2009), and Zhang et al. (2020) that maximise Sharpe or mean variance utility, the trading metrics naturally yield optimality conditions for calibrating the filter so that its statistical behaviour supports economically efficient trading decisions.

Paper structure

Section 2 completes mathematical analysis for the trading-oriented optimisation objective and its parameter selection to control optimisation behaviour. Section 3 investigates these empirical properties by modelling simulated regime changing data with CNN networks. Section 4 compares neural signal filters with state space neural networks. We provide an architecture-level comparison of the models' inference efficiency, structural interpretability, adaptability, and other structural characteristics relevant for trading. Section 5 construct a mean reversion trading control problem on the spread between spot price and filter signals. We compare signal filters by trading performances.

2 Trading-oriented Optimisation Objective

We discuss the optimisation objective and penalty terms to achieve suitable modelling characteristics for trading. Specifically, we show that LAFO, by matching average residuals within a moving window, dynamically captures moving average characteristics. The filtering window length K controls the low-pass kernel weighting. We examine trend filtering penalties to constrain the filter for both global smoothness and local responsiveness to different structural break assumptions. The total variation (TV) penalty gives piecewise-constant behaviour, $\ell 1$ trend filtering (TF) corresponds to piecewise-linear changes, and $\ell 2$ on first differences lead to global smoothness. We also show that the penalty restores solution uniqueness for rank-deficient filtering kernels.

2.1 Local average filtering objective (LAFO)

LAFO measures how well a filtered signal $\hat{y}$ matches the observed data y in terms of local average residuals (in contrast to pointwise fitting measures such as mean squared or mean absolute errors).

Meanwhile, this section formally defines LAFO, with appropriate boundary handling. We begin by writing the sample mean of residuals within a sliding window.

Definition 2.1 (Sliding window average)**.** For a window of length $K \geq 1$ at time index $t = 1, 2, ..., T$, we can define the causal window endpoints a_t, b_t and number of samples n_t in the window:

$$a_t := \max(1, t - K + 1), \quad b_t := t, \quad n_t := b_t - a_t + 1,$$

Now we can define the local average (within window) residual $\bar{e}_t$:

$$\bar{e}_t := \frac{1}{n_t} \sum_{k=a_t}^{b_t} (y_k - \hat{y}_k)$$

Definition 2.2 (LAFO)**.** LAFO is the average of the squared local average residuals over time T.

$$\mathcal{L}_{\text{LAFO}}(\hat{\mathbf{y}}) = \frac{1}{T} \sum_{t=1}^{T} \bar{e}_t^2 = \frac{1}{T} \sum_{t=1}^{T} (\frac{1}{n_t} \sum_{k=a_t}^{b_t} (y_k - \hat{y}_k))^2. \tag{2.1}$$

We can also write LAFO in the following quadratic form.

Definition 2.3 (LAFO as quadratic form)**.** Let $\mathbf{y}, \hat{\mathbf{y}} \in \mathbb{R}^T$, where $\mathbf{y}, \hat{\mathbf{y}}$ correspond to column vectors of the observed and fitted data. We have residuals $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$ and the causal sliding-average operator $A \in \mathbb{R}^{T \times T}$ (i.e., $\bar{e} = Ae = A(\mathbf{y} - \hat{\mathbf{y}})$), for fixed window length K (each window ends at t, with fixed-K padding). Define $W := A^\top A \succeq 0$, then

$$\mathcal{L}_{\text{LAFO}}(\hat{\mathbf{y}}) = \frac{1}{T} \|A(\mathbf{y} - \hat{\mathbf{y}})\|_2^2 = \frac{1}{T} (\mathbf{y} - \hat{\mathbf{y}})^\top W (\mathbf{y} - \hat{\mathbf{y}}) \tag{2.2}$$

LAFO is convex because $W \succeq 0$.

2.2 LAFO Properties

We show LAFO is a natural extension of the standard mean squared error (MSE, local pointwise fitting to high frequency noise). LAFO recovers the low-frequency or long-term average by having the adaptive window of size K, which controls the low pass kernel weighting, or the degree of filtering locality. LAFO becomes the standard MSE when with $K = 1$, and it retains only the global average when $K = T$. Any $1 < K < T$ is an interpolation between local fitting and global average. We show the influence of changing the window size on the nature of the matrix W in equation (2.2), leading to different convexity and uniqueness properties.

The following results hold at the extremes, with I being the identity matrix.

$$K = 1 \Rightarrow A = I \Rightarrow W = I \qquad \text{standard MSE, pointwise fit}$$

$$K = T \Rightarrow rank(W) = 1 \qquad \text{global mean-only fit, average of all residuals}$$

$K = 1$ **(pointwise fitting)** When $K = 1$, $\bar{e}_t := \frac{1}{n_t}\sum_{k=a_t}^{b_t}(y_k - \hat{y}_k) = I(y_t - \hat{y}_t)$ since $a_t = t, b_t = t, n_t = 1$. The sliding average operator A applies the identity operation:

$$(Ae)_t = e_t \Rightarrow A = I_T \qquad \text{(identity matrix of dimension } T \times T\text{)}$$

now with $W = A^\top A = I^\top I = I$:

$$\mathcal{L}_{\text{LAFO}}(\hat{\mathbf{y}}) = \frac{1}{T}\sum_{t=1}^{T}\bar{e}_t^2 = \frac{1}{T}(\mathbf{y} - \hat{\mathbf{y}})^\top I(\mathbf{y} - \hat{\mathbf{y}}) = \frac{1}{T}\|\mathbf{y} - \hat{\mathbf{y}}\|_2^2 \qquad \text{(standard MSE)}$$

The standard MSE is recovered by setting $K = 1$. For this objective, every individual (pointwise) error residual $y_t - \hat{y}_t$ directly influences the penalty and the model's training behaviour.

$K = T$ **(global averaging)** We will show in Fourier terms that LAFO becomes a zero-frequency (global constant) filter when the sliding window $K = T$. To enable Fourier analysis, we modify the causal boundary handling in Definition 2.1 to periodic padding. With causal boundaries, the averaging operator is Toeplitz: the window length varies at the edges, so the operator no longer represents a circular convolution. Toeplitz operators are not diagonalised by the DFT, and their Fourier behaviour cannot be expressed through a simple pointwise multiplier. Replacing the causal boundary with periodic padding removes this asymmetry. A circular convolution structure enables the use of the convolution theorem and the Fourier multiplier $\tilde{\varphi}_K(\omega_\ell)$.

Definition (Periodic indexing)**.** Let all signals be indexed by $t = 1, \ldots, T$. Define the periodic index map

$$\pi_T(k) := ((k-1) \bmod T) + 1,$$

which wraps any integer index into $\{1, \ldots, T\}$.

Definition (Periodic sliding-average operator)**.** Let $e_t = y_t - \hat{y}_t$ denote the residuals. For a window length K, define the boxcar kernel $\varphi_K \in \mathbb{R}^T$ by

$$\varphi_K[r] = \begin{cases} \frac{1}{K}, & r = 1, \ldots, K, \\ 0, & r = K+1, \ldots, T. \end{cases}$$

so that the circular moving average is

$$(Ae)_t = (\varphi_K * e)_t = \sum_{m=0}^{T-1}\varphi_K[m+1]\, e_{\pi_T(t-m)} = \frac{1}{K}\sum_{j=1}^{K} e_{\pi_T(t-j+1)}.$$

The convolution matrix entries are

$$A_{t,k} = \varphi_K(\pi_T(t-k)+1).$$

For $t = 1$, the nonzero columns are

$$\pi_T(1),\ \pi_T(0),\ \pi_T(-1), \ldots, \pi_T(2-K),$$

that is,

$$1,\ T,\ T-1, \ldots, T-K+2.$$

Hence the first row of A is

$$A_{1,:} = \big(1/K,\ 0,\ 0, \ldots, 0,\ 1/K,\ 1/K, \ldots, 1/K\big),$$

consisting of one $1/K$ at position 1 and $K-1$ entries of $1/K$ at the end of the row. Each subsequent row is a cyclic shift of the previous row, so A is circulant under periodic padding.

Definition (Discrete Fourier transform)**.** For a signal x_t, $t = 1, \ldots, T$, define the discrete Fourier transform (DFT) by

$$\tilde{x}(\omega_\ell) = \frac{1}{\sqrt{T}} \sum_{t=1}^{T} x_t e^{-i\omega_\ell(t-1)}, \qquad \omega_\ell = \frac{2\pi\ell}{T}, \quad \ell = 0, \ldots, T-1.$$

The inverse transform is

$$x_t = \frac{1}{\sqrt{T}} \sum_{\ell=0}^{T-1} \tilde{x}(\omega_\ell) e^{i\omega_\ell(t-1)}.$$

Proposition (Spectral form of LAFO)**.** Under periodic padding, the LAFO objective admits the equivalent spectral representation

$$\mathcal{L}_{\mathrm{LAFO}}(\hat{\mathbf{y}}) = \sum_{\ell=0}^{T-1} |\tilde{\varphi}_K(\omega_\ell)|^2 |\tilde{e}(\omega_\ell)|^2.$$

Proof. To characterise the LAFO objective, we transform its quadratic form

$$\mathcal{L}_{\mathrm{LAFO}}(\hat{\mathbf{y}}) = \frac{1}{T}\|Ae\|_2^2, \qquad e = y - \hat{y},$$

to Fourier domain. Matrix A is circulant and therefore diagonalised by the DFT. Applying the DFT to Ae and using the convolution theorem yields

$$Ae = \varphi_K * e, \qquad \widetilde{Ae}(\omega_\ell) = \sqrt{T}\tilde{\varphi}_K(\omega_\ell)\tilde{e}(\omega_\ell).$$

By Plancherel's theorem,

$$\|Ae\|_2^2 = \|\widetilde{Ae}\|_2^2 = \sum_{\ell=0}^{T-1} |\widetilde{Ae}(\omega_\ell)|^2.$$

Substituting the convolution identity $\widetilde{Ae}(\omega_\ell) = \sqrt{T}\,\tilde{\varphi}_K(\omega_\ell)\,\tilde{e}(\omega_\ell)$,

$$\|Ae\|_2^2 = \sum_{\ell=0}^{T-1} |\sqrt{T}\,\tilde{\varphi}_K(\omega_\ell)\,\tilde{e}(\omega_\ell)|^2 = T\sum_{\ell=0}^{T-1} |\tilde{\varphi}_K(\omega_\ell)|^2\,|\tilde{e}(\omega_\ell)|^2.$$

Now, we have

$$\mathcal{L}_{\text{LAFO}} = \frac{1}{T}\|Ae\|_2^2 = \sum_{\ell=0}^{T-1} |\tilde{\varphi}_K(\omega_\ell)|^2\,|\tilde{e}(\omega_\ell)|^2,$$

Proposition (Zero-frequency filtering for $K = T$)**.** When $K = T$,

$$\mathcal{L}_{\text{LAFO}}(\hat{\mathbf{y}}) = \frac{1}{T}|\tilde{e}(\omega_0)|^2 = \frac{1}{T^2}\left(\sum_{t=1}^{T} e_t\right)^2.$$

Proof. Starting from the Spectral form of LAFO, first compute the DFT $\tilde{\varphi}_K(\omega)$, recall that the boxcar has value $1/K$ for $t = 0, \ldots, K-1$ and zero otherwise,

$$\tilde{\varphi}_K(\omega_\ell) = \frac{1}{\sqrt{T}} \cdot \frac{1}{K}\sum_{t=0}^{K-1} e^{-i\omega_\ell t}.$$

For $\ell \neq 0$, the sum forms a finite geometric series:

$$\sum_{t=0}^{K-1} r^t = \frac{1-r^K}{1-r}, \qquad r = e^{-i\omega_\ell},$$

we obtain

$$\tilde{\varphi}_K(\omega_\ell) = \frac{1}{\sqrt{T}} \cdot \frac{1}{K}\frac{1-e^{-i\omega_\ell K}}{1-e^{-i\omega_\ell}},$$

Setting $K = T$, the numerator becomes

$$1 - e^{-i\omega_\ell T} = 1 - e^{-i(2\pi\ell/T)T} = 1 - e^{-i2\pi\ell}.$$

$e^{-i2\pi\ell} = 1$ and $1 - e^{-i\omega_\ell} \neq 0$ for $\ell = 1, \ldots, T-1$, it follows that

$$\tilde{\varphi}_T(\omega_\ell) = 0, \qquad \ell = 1, \ldots, T-1.$$

For $\ell = 0$, we have $r = e^{-i\omega_0} = 1$, and the geometric-series formula yields $0/0$. In this case, we

return to the definition of the sum:

$$\sum_{t=0}^{T-1} r^t = \sum_{t=0}^{T-1} 1 = T.$$

Therefore,

$$\tilde{\varphi}_T(0) = \frac{1}{\sqrt{T}} \cdot \frac{1}{T} \cdot T = \frac{1}{\sqrt{T}}.$$

Consequently,

$$|\tilde{\varphi}_T(\omega_\ell)|^2 = \begin{cases} 1/T, & \ell = 0, \\ 0, & \ell = 1, \ldots, T-1. \end{cases}$$

Since $|\tilde{\varphi}_T(\omega_\ell)|^2 = 0$ for $\ell \neq 0$ and $|\tilde{\varphi}_T(\omega_0)|^2 = 1/T$, only the DC term remains:

$$\mathcal{L}_{\text{LAFO}} = \frac{1}{T} |\tilde{e}(\omega_0)|^2.$$

Then, we compute $\tilde{e}(\omega_0)$. By the definition of DFT,

$$\tilde{e}(\omega_0) = \frac{1}{\sqrt{T}} \sum_{t=0}^{T-1} e[t],$$

hence

$$\mathcal{L}_{\text{LAFO}} = \frac{1}{T} \cdot \frac{1}{T} \left| \sum_{t=0}^{T-1} e[t] \right|^2 = \frac{1}{T^2} \left(\mathbf{1}^\top (y - \hat{y}) \right)^2.$$

In this limit, A retains only the zero-frequency component of the residuals, and

$$W = A^\top A = \frac{1}{T} \mathbf{1}\mathbf{1}^\top,$$

W has rank one and $\ker(W) = \mathbf{1}^\perp = \{\mathbf{z} : \mathbf{1}^\top \mathbf{z} = 0\}$.

Connections to Trading With a small trading window, when $K \to 1$, the objective leads to overfitting to short-term noise, obscuring the low-frequency fair price signals. Meanwhile, the global average solution when $K \to T$ is static, unresponsive, and ignorant of local price series characteristics such as time-varying trends and regime changes.

In contrast, a suitable selection $1 < K < T$ ensures that matrix W maintains its rank and gives LAFO a unique minimum. The resulting model will have a suitable balance between local fit and robustness to long-term noise. The relationship is summarised by the following table:

Table 1: Comparison of LAFO behavior across window lengths.

Property	$\mathbf{K = 1}$	$\mathbf{1 < K < T}$	$\mathbf{K = T}$
Matrix W	I	Banded Toeplitz	$\frac{1}{T}\mathbf{1}\mathbf{1}^\top$
rank(W)	T	T (typically)	1
ker(W)	$\{0\}$	$\{0\}$	$\mathbf{1}^\perp$ (dim $T-1$)
Objective reduces to	MSE	Sliding avg. of residuals	Global mean only
Uniqueness	Guaranteed	Likely	**Fails**
Trading suitability	Poor (overfit)	**Good**	Poor (constant signal)

2.3 Regime Adaptive Penalty

The penalty term encodes structural assumptions about regime changes through global smoothness (LAFO-$\ell 2$), piecewise-constant (LAFO-TV), or piecewise-linear (LAFO-$\ell 1$ structural constraints. The Rudin et al. (1992) Total variation (TV) penalty and S.-J. Kim et al. (2009) ℓ_1 trend filtering objectives apply ℓ_1 penalties on the first and second differences of the model output. Tibshirani (2014) shows that the TV penalty promotes piecewise-constant structure, while the ℓ_1 trend filtering objective is favourable on piecewise-linear structure with sparse, data-driven jumps. Lastly, Tikhonov and Arsenin (1977) and Eilers (2003) ℓ_2 regularisation uses squared first differences, enforcing global smoothness. The ℓ_1-based penalties (TV and trend filtering) better preserve abrupt regime changes and reduce lag relative to Tikhonov; related ℓ_1 filters have been applied to momentum signals primarily with MSE losses in Dao (2014).

Definition 2.4 (LAFO-TV)**.** ℓ_1 norm on first differences. This corresponds to level-stationary regimes in which the latent fair value remains approximately constant over time, with infrequent discrete structural breaks.

$$\mathcal{L}_{\text{LAFO-TV}} = \mathcal{L}_{\text{LAFO}}(\hat{\mathbf{y}}) + \lambda_{\text{TV}} \sum_{t=2}^{T} |\hat{y}_t - \hat{y}_{t-1}|. \tag{2.3}$$

Definition 2.5 (LAFO-ℓ_1 TF)**.** ℓ_1 norm on second differences. This encodes trend-stationary regimes in which the latent fair value evolves linearly within regimes, with sparse and abrupt changes in slope corresponding to regime transitions.

$$\mathcal{L}_{\text{LAFO-}\ell_1} = \mathcal{L}_{\text{LAFO}}(\hat{\mathbf{y}}) + \lambda_{\ell_1} \sum_{t=3}^{T} |\hat{y}_t - 2\hat{y}_{t-1} + \hat{y}_{t-2}|. \tag{2.4}$$

Definition 2.6 (LAFO-ℓ_2)**.** ℓ_2 norm on first differences. This encodes trend-stationary regimes in which the latent fair value evolves smoothly over time, with changes in slope occurring continuously rather than through discrete structural breaks. This corresponds to a diffusive evolution of the trend

component, analogous to a local-level or local-trend state-space model with Gaussian innovations.

$$\mathcal{L}_{\ell_2\text{-LAFO}} = \mathcal{L}_{\text{LAFO}}(\hat{\mathbf{y}}) + \lambda_{\ell_2} \sum_{t=2}^{T} (\hat{y}_t - \hat{y}_{t-1})^2 . \tag{2.5}$$

2.4 Closed Form Solution of Penalised LAFO and Convexity, Uniqueness

We showed in section 2.2 that LAFO without penalty gives infinitely many solutions when $K \sim T$. Large window size K makes the kernel W rank-deficient, creating a poorly-conditioned, numerically unstable optimisation problem. We will show that LAFO with ℓ_2 penalty gives a closed-form solution for a linear symmetric positive definite system, which recovers convexity and uniqueness. We first analyse the unstructured case $\hat{\mathbf{y}} \in \mathbb{R}^T$, then the structured case $\hat{\mathbf{y}} = X\boldsymbol{\beta}$. In contrast, LAFO with ℓ_1 penalties (TV/TF) is still convex but requires iterative optimisation.

Unstructured LAFO-ℓ_2. With $D^{(k)}$ being the difference operator of order k, the penalised estimator solves

$$\arg\min_{\hat{\mathbf{y}} \in \mathbb{R}^T} \frac{1}{T}(\mathbf{y} - \hat{\mathbf{y}})^\top W(\mathbf{y} - \hat{\mathbf{y}}) + \lambda \|D^{(k)}\hat{\mathbf{y}}\|_2^2, \qquad \lambda > 0.$$

The gradient is $\nabla_{\hat{\mathbf{y}}}\mathcal{L}(\hat{\mathbf{y}}) = \frac{2}{T}W(\hat{\mathbf{y}} - \mathbf{y}) + 2\lambda(D^{(k)})^\top D^{(k)}\hat{\mathbf{y}}$ and the Hessian is

$$H = \frac{1}{T}W + \lambda(D^{(k)})^\top D^{(k)}.$$

Both terms are positive semidefinite. For any $\mathbf{z}$,

$$\mathbf{z}^\top H\mathbf{z} = \frac{1}{T}\|W^{1/2}\mathbf{z}\|_2^2 + \lambda\|D^{(k)}\mathbf{z}\|_2^2,$$

which equals zero iff $W\mathbf{z} = 0$ and $D^{(k)}\mathbf{z} = 0$, i.e. $\mathbf{z} \in \ker(W) \cap \ker(D^{(k)})$. Thus

$$H \succ 0 \quad \Longleftrightarrow \quad \ker(W) \cap \ker(D^{(k)}) = \{\mathbf{0}\}.$$

Since a convex quadratic has a unique minimiser exactly when its Hessian is positive definite, the unstructured estimator is unique under this condition. The closed-form solution is

$$\hat{\mathbf{y}}^* = \left(\frac{1}{T}W + \lambda(D^{(k)})^\top D^{(k)}\right)^{-1} \frac{1}{T}W\mathbf{y}.$$

Structured LAFO-ℓ_2 with $\hat{\mathbf{y}} = X\boldsymbol{\beta}$. When $\hat{\mathbf{y}}$ has structure $X\boldsymbol{\beta}$, and $X \in \mathbb{R}^{T\times d}$ is a model matrix, constructed from basis functions such as regressions, splines, Fourier terms, or lagged values. The objective becomes

$$\mathcal{L}(\boldsymbol{\beta}) = \frac{1}{T}(\mathbf{y} - X\boldsymbol{\beta})^\top W(\mathbf{y} - X\boldsymbol{\beta}) + \lambda\|D^{(k)}X\boldsymbol{\beta}\|_2^2.$$

Expanding yields a quadratic form with gradient

$$\nabla_{\boldsymbol{\beta}}\mathcal{L}(\boldsymbol{\beta}) = \frac{2}{T}X^\top W X\boldsymbol{\beta} - \frac{2}{T}X^\top W\mathbf{y} + 2\lambda X^\top D^\top D X\boldsymbol{\beta},$$

and Hessian

$$H_{\boldsymbol{\beta}} = \frac{1}{T}X^\top W X + \lambda X^\top D^\top D X.$$

For any $\mathbf{z}$,

$$\mathbf{z}^\top H_{\boldsymbol{\beta}}\mathbf{z} = \frac{1}{T}\|W^{1/2}X\mathbf{z}\|_2^2 + \lambda\|DX\mathbf{z}\|_2^2,$$

this expression is 0 when $W(X\mathbf{z}) = 0$ and $D^{(k)}(X\mathbf{z}) = 0$, that is, when $X\mathbf{z} \in \ker(W) \cap \ker(D^{(k)})$. Since every vector of the form $X\mathbf{z}$ lies in $\mathrm{col}(X)$, the uniqueness condition is

$$H_{\boldsymbol{\beta}} \succ 0 \quad \Longleftrightarrow \quad \mathrm{col}(X) \cap \ker(W) \cap \ker(D^{(k)}) = \{\mathbf{0}\}.$$

The closed-form estimator is

$$\boldsymbol{\beta}^* = \left(\frac{1}{T}X^\top W X + \lambda X^\top D^\top D X\right)^{-1}\frac{1}{T}X^\top W\mathbf{y}, \qquad \hat{\mathbf{y}}^* = X\boldsymbol{\beta}^*.$$

LAFO-ℓ_1 penalties (TV/TF). Replacing the quadratic smoothness penalty with an ℓ_1 term gives the objective

$$\arg\min_{\hat{\mathbf{y}}\in\mathbb{R}^T}\frac{1}{T}(\mathbf{y}-\hat{\mathbf{y}})^\top W(\mathbf{y}-\hat{\mathbf{y}}) + \lambda\|D^{(k)}\hat{\mathbf{y}}\|_1, \qquad \lambda > 0.$$

The first term is a convex quadratic, and the second term is an ℓ_1 norm of a linear transform, so the problem remains convex. However, the ℓ_1 penalty is non-differentiable whenever $(D^{(k)}\hat{\mathbf{y}})_i = 0$. The quadratic plus ℓ_1 objective is convex but non-smooth. Solutions can be obtained by iterative optimisation algorithms such as proximal-gradient, ADMM, or primal–dual methods.

3 Regime Adaptive Simulation on LAFO

We demonstrate the empirical properties of LAFO, with and without penalty, through simulations. The first section introduces the regime-adaptive filtering simulation protocol. We rationalise the usage of a simple 2-layer CNN model to show interactions between neural network models and LAFO in adaptive filtering tasks. We firstly show the influence of different LAFO window length K on the filtering behaviour. We also compare different penalty terms on the filtered signal.

3.1 Regime-adaptive filtering simulation protocol

We simulate a piecewise TrendARMA process: within each regime $r = 1, \ldots, R$, the deviation x_t follows an ARMA(p, q) driven by heavy-tailed or mixture innovations, while the mean evolves

linearly. The observed series is their sum:

$$x_t = \sum_{i=1}^{p} \phi_i^{(r)} x_{t-i} + \sum_{j=1}^{q} \theta_j^{(r)} \varepsilon_{t-j} + \varepsilon_t, \qquad \mu_t = \beta_0^{(r)} + \beta_1^{(r)} t, \qquad y_t = \mu_t + x_t. \tag{3.1}$$

The ARMA model x_t is stationary, consisting of a low-frequency predictable signal and a white noise component ε_t. Regime boundaries permit intercept shifts and slope changes through $(\beta_0^{(r)}, \beta_1^{(r)})$. AR/MA coefficients may also change. Innovations ε_t are sampled from Student-t (df, scale) or Gaussian mixtures (weights, means, stds) for tail risk and mild volatility clustering.

3.2 Structural model assumptions

We train a minimal 2-layer CNN architecture, varying the training objectives, to validate the usefulness of each objective. The first layer captures multiple ARMA-type temporal dependencies, and the second layer adaptively combines them across regimes:

$$\hat{y}_t = \sum_{c=1}^{C} (v_c \odot \sigma(w_c \odot x))_t, \tag{3.2}$$

where $(w_c * x)_t = \sum_{\ell=0}^{K-1} w_{c,\ell} x_{t-\ell}$ represents an ARMA-like convolutional filter, $\sigma(\cdot)$ is a nonlinear activation (e.g., ReLU) applied between layers, and v_c are second-layer convolutional weights that control the selection or combination of these filters over time. Here, C denotes the number of first-layer convolutional channels, each corresponding to an implicit ARMA filter capturing a distinct regime-dependent dynamic. The inclusion of the nonlinear activation allows the second layer to perform nonlinear mixing of ARMA filters, so the model can approximate switching or state-dependent behaviour. The first layer of the CNN has a convolution kernel of size 512 samples, giving access to long-range information (2-4 regimes in simulated datasets). The second layer (kernel size 1) only mixes across channels of different filtering kernels. M. J. Kim et al. (2025) present a similar approach using CNN models to capture ARMA characteristics, trained based on MAE and MSE for forecasting.

3.3 LAFO and sliding window length

We show the impact of different LAFO window lengths K on the neural network filter. There are several possibilities to select K in a data-informed way. One option is to choose the window length relative to expected regime duration, for example setting it near the median estimated regime half-life so that LAFO accentuates deviations on that horizon. A second approach uses the autocorrelation of returns, choosing K as the smallest lag at which $|\rho_{\Delta y}(\tau)|$ falls below a threshold (e.g., 0.1). Finally, K may be selected adaptively via rolling walk-forward cross-validation on a proxy criterion such as out-of-sample low-frequency MSE or a trading-utility measure, using a discrete grid $K \in \{8, 16, 32, 64, 128, \dots\}$. These heuristics give practical guidance for setting K

depending on the application and the characteristic time scales of the data. Figure 1 shows the window size controls the low-pass weighting. Larger K corresponds to lower frequency filters, while a model with $K = 1$ retains all high-frequency noise.

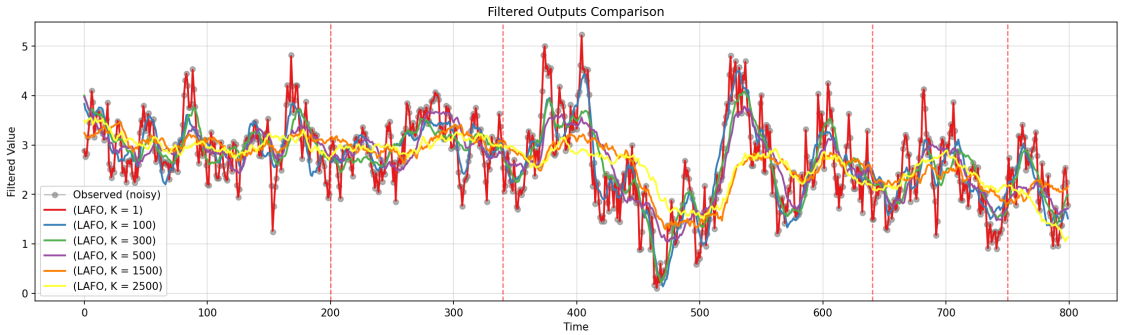


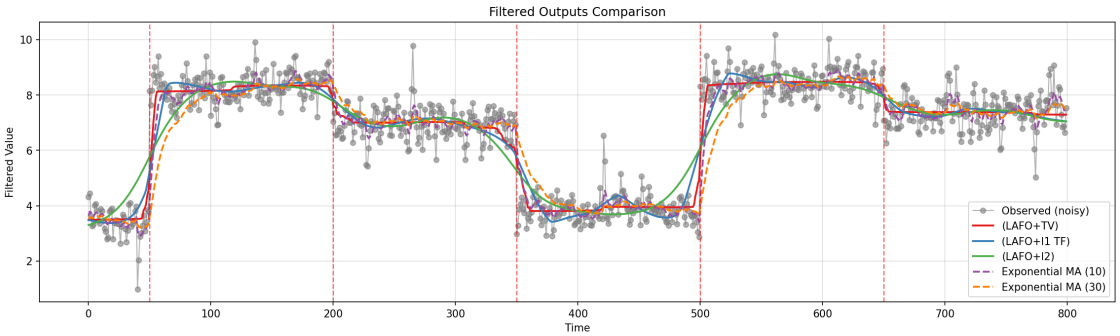
Figure 1: LAFO with different sliding window length, trained on 2-layer simple CNN model. The figures show the last 800 samples of the datasets. Red dashed vertical lines show regime boundaries.

3.4 Penalised LAFO

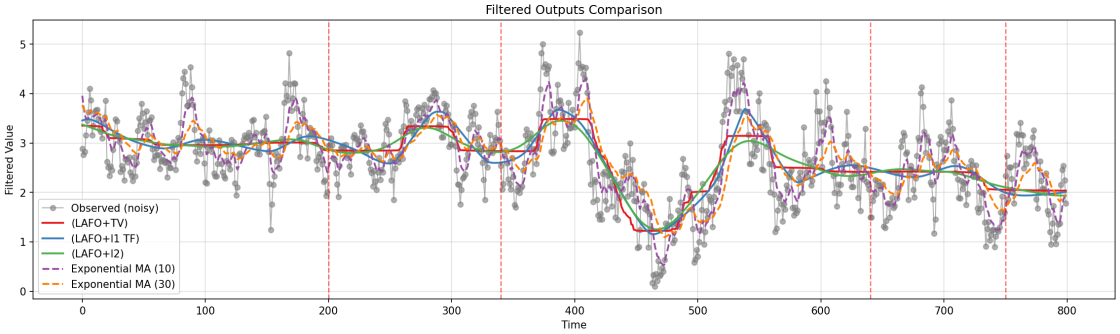
We compare LAFO-TV, LAFO-ℓ_1 TF, and LAFO-ℓ_2. Following the approach of S.-J. Kim et al. (2009), model penalties λ are calibrated so the resulting models achieve similar training losses. We evaluate models against exponential moving average (EMA) baseline models at two different lags, 10 and 30. The EMA(10) baseline works as a high-frequency filter, whereas EMA(30) retains primarily low-frequency structure. Ideally, the filtered signals should exhibit the global smoothness characteristic of EMA(30) while remaining responsive to structural changes, as EMA(10) does.

Due to the CNN structural constraints, using the same K yields similar filter weightings for all models, but different penalties induce distinct balances between smoothness and adaptability to changes in level or trend. We evaluate the models on datasets of two different natures: piecewise-linear and piecewise-constant trends. While their ARMA models x_t remain the same, the piecewise-constant data does not have a $\beta_1^{(r)} t$ component in its μ_t (see equation (3.1) for model specification).

We choose $K = 80$. In simulated datasets, regime change occurs every 100 to 300 samples. A large K (and large smoothing penalty) leads to pooling across more than 2 different regimes. This is undesirable unless regime change is slow or semi-Markovian. Figure 2 shows how the penalties preserve low-frequency structure and remain responsive to abrupt changes in regimes. The fitting results confirm the analyses in S.-J. Kim et al. (2009) and Tibshirani (2014): LAFO-ℓ_2 promotes overall smoothness, LAFO-ℓ_1 shows piecewise-constant trends, and LAFO-ℓ_1 TF displays piecewise-linear characteristics.



(a) Piecewise constant



(b) Piecewise linear

Figure 2: Penalised LAFO, $K = 80$, trained on 2-layer simple CNN model. The figures show the last 800 samples of the datasets. Red dashed vertical lines show regime boundaries.

4 Advanced Filtering Implementations

We treat neural networks as flexible parameterisations of filtering operators whose behaviour is governed by the underlying objective. Within the LAFO framework, different architectures correspond to different constraints on memory, adaptivity, and regime responsiveness.

The CNN model has access to richer context (historical information) to help its inference compared to moving average models, but simple CNN model convolution kernels are unstructured. This is ineffective since financial time series are weakly autocorrelated. In contrast, advanced neural network architectures incorporate novel mechanisms for long-range information retrieval and nonlinear sequence modelling. Filtering models may be categorised by whether they are based on or influenced by state-space and Bayesian estimation methods. We produce a comprehensive qualitative table that compares the strengths and limitations of the models based on key structural properties, including memory horizon, inference efficiency, data requirements, adaptability to different noise and signal structures, and other relevant factors.

4.1 Models without state space formulation

The Wavenet model in Oord et al. (2016) uses dilated causal convolutions with activations to create a nonlinear finite impulse response (FIR) style filter with learnable weights. The Dual-Path RNN (DPRNN) model by Luo et al. (2020) applies RNN modelling on the long and short range to capture both local temporal context and global dependencies. The Informer model by H. Zhou et al. (2021) uses Transformers with sparse attention to model the decomposed trend and residuals.

4.1.1 Wavenet

The WaveNet model (Oord et al., 2016) introduces a stack of dilated causal convolutions to capture long-range dependencies with fixed-size filters. The key operation at layer l is defined by:

$$h_t^{(l)} = \sum_{i=0}^{k-1} w_i^{(l)} h_{t-i\cdot d_l}^{(l-1)}, \tag{4.1}$$

$h_t^{(l)}$ is the hidden activation at time t, k is the kernel size, d_l is the dilation factor at layer l, and $w_i^{(l)}$ are learnable weights. By setting $d_l = 2^l$, the receptive field expands exponentially with depth.

A gated activation follows each convolution:

$$z_t = \tanh(W_f * h^{(l-1)}) \odot \sigma(W_g * h^{(l-1)}) \tag{4.2}$$

where W_f and W_g are convolutional filters, and $\odot$ denotes element-wise multiplication. This structure allows the model to behave as an adaptive ensemble of input-conditioned FIR filters. Temporal Convolutional Networks (TCN) by Lea et al. (2017) extends the architecture with residual connections, layer normalization, and stable training mechanisms, making TCNs more robust for supervised sequence learning tasks.

4.1.2 Informer

H. Zhou et al. (2021) develop a Transformer-based model for long-sequence forecasting and filtering. It introduces a sparse attention mechanism, termed ProbSparse attention:

$$\mathrm{Attn}_{\mathrm{prob}}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} \odot M\right) V, \tag{4.3}$$

where $M \in \{0,1\}^{L\times L}$ masks all but the top-u queries selected by a sparsity score (based on the divergence of each query's attention from the uniform distribution). This yields sub-quadratic complexity $O(L \log L)$ while preserving access to long-range dependencies. Autoformer from Wu et al. (2022) is an extension of Informer that models both trend and residual components using a recursive series decomposition, where the trend is propagated through skip connections and exhibits

an autocorrelated structure conducive to long-horizon forecasting. T. Zhou et al. (2022) further extends this to frequency-domain cross-attention with Fourier transform in FEDformer.

4.1.3 Dual-Path RNN

Luo et al. (2020) segments a length-L sequence $X \in \mathbb{R}^{L\times d}$ into C overlapping chunks of length K with stride S via a chunking operator $\mathcal{C}_{K,S}$, where $(\mathcal{C}_{K,S}(X))_{c,k,:} = X_{cS+k,:}$ for $c = 0, \dots, C-1$, $k = 0, \dots, K-1$ and $C = \lfloor (L-K)/S \rfloor + 1$. Each block applies an intra-chunk RNN along the chunk axis and an inter-chunk RNN across chunks:

$$H^{\text{intra}}_{c,:,:} = \text{RNN}_{\text{intra}}(X_{c,:,:}), \qquad H^{\text{inter}}_{:,k,:} = \text{RNN}_{\text{inter}}(H^{\text{intra}}_{:,k,:}), \tag{4.4}$$

then projects and reconstructs the sequence with an overlap–add operator $\mathcal{O}_{K,S}$ that sums overlapping frames and averages by their multiplicity:

$$Y = \mathcal{O}_{K,S}\big(\text{Proj}(H^{\text{intra}} + H^{\text{inter}})\big) \in \mathbb{R}^{L\times d}. \tag{4.5}$$

Stacking dual-path blocks yields a global receptive field with $O(L)$ sequential cost. Structurally, DPRNN sits between WaveNet and Informer: it extends fixed local convolutions by routing information across chunks (serving a role similar to attention for long-range exchange) while retaining recurrent inductive bias and sequential computation. Chen et al. (2020) replace the RNNs with Transformers to strengthen nonlocal matching at higher cost.

4.2 Neural state space models

Assuming that x_t is the unobservable, hidden, or latent state vector at time t, a general (possibly nonlinear) discrete-time state space model is written as:

$$\begin{aligned} x_t &= f_t(x_{t-1}, w_t), \quad w_t \sim \mathcal{D}_t \\ y_t &= g_t(x_t, v_t), \quad v_t \sim \mathcal{E}_t \end{aligned} \tag{4.6}$$

y_t is the observed measurement. The state transition function f_t governs the evolution of the hidden state, and the observation or emission function g_t describes how the hidden state maps to the observed output. The random variables w_t and v_t represent process and observation noise, drawn from generic distributions $\mathcal{D}_t$ and $\mathcal{E}_t$, respectively.

Deep Kalman Filters (DKFs) combine the Bayesian inference framework with deep learning to estimate the latent state in nonlinear, non-Gaussian systems. A. Gu et al. (2022) develops the Structured State Space Models (S4) model, which leverages a convolutional representation of the state space solution to enable efficient long-range sequence modelling in linear time and space complexity.

4.2.1 Deep Kalman Filters

Krishnan et al. (2015) extends the classical Kalman filter to nonlinear settings by parametrising the state transition and emission distributions with neural networks f_{trans} and f_{emit} with $\theta = \{\theta_{\text{trans}}, \theta_{\text{emit}}\}$, the generative parameters.:

$$\begin{aligned} x_t &\sim \mathcal{N}(\mu_t^{(x)}, \Sigma_t^{(x)}), \quad (\mu_t^{(x)}, \Sigma_t^{(x)}) = f_{\text{trans}}(x_{t-1}; \theta_{\text{trans}}), \\ y_t &\sim \mathcal{N}(\mu_t^{(y)}, \Sigma_t^{(y)}), \quad (\mu_t^{(y)}, \Sigma_t^{(y)}) = f_{\text{emit}}(x_t; \theta_{\text{emit}}), \end{aligned} \tag{4.7}$$

Inference at each time step mirrors the structure of classical Bayesian filtering. Since the transition and emission functions are nonlinear, the posterior $p(x_t \mid y_{1:t})$ is generally intractable. DKF introduces a variational inference network f_{enc} to approximate the posterior distribution, where ϕ denotes the parameters of the encoder:

$$q_\phi(x_t \mid y_{1:t}) = \mathcal{N}(\mu_t^{(q)}, \Sigma_t^{(q)}), \quad (\mu_t^{(q)}, \Sigma_t^{(q)}) = f_{\text{enc}}(y_{1:t}; \phi), \tag{4.8}$$

Conditional Prior. A prior distribution over x_t is computed from the previous latent state using the transition model:

$$p_\theta(x_t \mid x_{t-1}) = \mathcal{N}(\mu_t^{(x)}, \Sigma_t^{(x)}). \tag{4.9}$$

Bayesian Update and Output. The inference network conditions on all past observations $y_{1:t}$ to approximate the posterior over x_t. The predicted observation is then decoded from this posterior:

$$\hat{y}_t = \mathbb{E}_{q_\phi(x_t|y_{1:t})}[f_{\text{emit}}(x_t)] \approx f_{\text{emit}}(\mu_t^{(q)}). \tag{4.10}$$

The model parameters θ and variational parameters ϕ are learned by maximizing the evidence lower bound (ELBO) on the marginal log-likelihood $\log p_\theta(y_{1:T})$. For each time step t, the ELBO decomposes into a data fit term and a regularization term:

$$\mathcal{L}_{\text{ELBO}}^{(t)} = \mathbb{E}_{q_\phi(x_t|y_{1:t})}\big[\log p_\theta(y_t \mid x_t)\big] - \text{KL}(q_\phi(x_t \mid y_{1:t}) \parallel p_\theta(x_t \mid x_{t-1}))\,. \tag{4.11}$$

The first term is the expected log-likelihood, which encourages the inferred latent state x_t to explain the observed data y_t through the emission model. Under the Gaussian emission assumption,

$$p_\theta(y_t \mid x_t) = \mathcal{N}\big(y_t \mid \mu_t^{(y)}, \Sigma_t^{(y)}\big), \qquad (\mu_t^{(y)}, \Sigma_t^{(y)}) = f_{\text{emit}}(x_t),$$

this term reduces to a weighted reconstruction error in observation space.

The second term is the Kullback-Leibler divergence between the approximate posterior and the conditional prior,

$$\text{KL}(q_\phi(x_t \mid y_{1:t}) \parallel p_\theta(x_t \mid x_{t-1}))\,,$$

which regularises the inference network by penalising deviations from the dynamics implied by the

transition model. This term enforces temporal consistency of the latent state sequence and prevents overfitting to individual observations.

Summing over time yields the full training objective:

$$\mathcal{L}_{\text{ELBO}} = \sum_{t=1}^{T} \mathbb{E}_{q_\phi(x_t|y_{1:t})}\big[\log p_\theta(y_t \mid x_t)\big] - \sum_{t=1}^{T} \mathrm{KL}(q_\phi(x_t \mid y_{1:t}) \parallel p_\theta(x_t \mid x_{t-1}))\,. \tag{4.12}$$

The original DKF assumes Gaussian transition, emission, and inference distributions to enable closed-form training objectives. Maddison et al. (2017) propose a particle filter based variational bound that relaxes the Gaussian assumptions. Nguyen-Quynh et al. (2021) extends the classical switching linear state-space model framework of Mesot and Barber (2007) by introducing a neural switching Kalman filter. A discrete regime variable s_t gates regime-specific nonlinear transition and emission networks, yielding interpretable regime segmentation and capturing multimodality, abrupt shifts, and regime-dependent volatility. However, the explicit switching layer increases nonconvexity and sample complexity, risks label-switching and training instability.

4.2.2 Structured State Space Sequence Model (S4)

A. Gu et al. (2022) achieves greater efficiency than conventional RNNs and Transformers by exploiting a structured transition matrix and fast kernel application. The model admits three equivalent formulations: a discrete-time view aligned with the discrete nature of financial data and exposing eigenvalues that determine memory timescales; a convolutional view that enables parallel training; a recurrent view supporting constant-latency online inference.

Discrete view. Where $u_t \in \mathbb{R}^{d_u}$ is the input, $y_t \in \mathbb{R}^{d_y}$ is the output, state $x_t \in \mathbb{C}^N$ follows:

$$x_{t+1} = \bar{A}x_t + \bar{B}u_t, \qquad y_t = Cx_t + Du_t, \tag{4.13}$$

In our setting, the input u_t corresponds directly to the observed noisy signal (e.g., traded price), and the output y_t is the filtered fair value estimate. This makes the model a single-input, single-output filter applied to the observed series.

Recurrent view. Unrolling the recurrence on internal state x_t, assuming zero initial state $x_0 = 0$, the state and output are given by:

$$x_t = \bar{A}^t x_0 + \sum_{\tau=0}^{t-1} \bar{A}^{t-1-\tau}\bar{B}u_\tau, \qquad y_t = Cx_t + Du_t = \sum_{\tau=0}^{t-1} C\bar{A}^{t-1-\tau}\bar{B}u_\tau + Du_t. \tag{4.14}$$

Convolutional view. The linear recurrence defines a time-invariant system, whose output depends on a weighted sum of past inputs. This allows the system to be equivalently expressed as a causal

convolution. The full kernel $h_{0:L}$ can be computed and applied via FFT in $O(L \log L)$, enabling fast parallel training on long sequences:

$$y = h * u, \qquad h_t = C\bar{A}^{t-1}\bar{B} + \delta_{t0}D, \tag{4.15}$$

With $\bar{A} := \mathrm{diag}(\lambda) + UV^\top$, the diagonal part defines independent exponential decays, and $\mathrm{Re}(\lambda_i) < 0$ ensures stability. The low-rank term $UV^\top$ introduces cross-coupling between latent dimensions. This expands the expressiveness of the kernel beyond simple exponential smoothing. A. Gu et al. (2020) follows the HiPPO initialization for $\bar{A}$ to encode long-horizon memory by projecting recent inputs onto a basis of orthogonal polynomials stored in the state. The Mamba model A. Gu and Dao (2024) generalizes this architecture by removing HiPPO and making B and C input-dependent, enabling content-based memory access without explicitly computing a distinct kernel at each time step.

4.3 Model Comparison

We show in Table 2 each model's key structural characteristic and evaluate their advantages across multiple criteria relevant to fair price filtering and mean-reversion trading, including memory dynamics, context adaptivity, training efficiency, latency, inference stability, interpretability, and sample efficiency under low signal-to-noise conditions.

Table 2: **Comparison of neural sequence models for fair price filtering in mean-reversion trading.** CNN-based filters such as Wavenet achieve long memory through fixed receptive fields defined by kernel sizes and dilation schedules. RNN-based models like DPRNN extend memory through recurrent states, allowing content-adaptive context but at the cost of sequential processing. Transformer-based models such as Informer employ attention to achieve global, variable-length memory with full adaptivity, but typically require more data and have higher computational overhead. State-space models, including DKF and S4, encode history into compact latent states, achieving theoretically infinite or very long memory with efficient inference.

Model	Memory Dynamics	Context Adaptivity	Training Efficiency	Per-Step Latency	Inference Stability	Sample Efficiency (Low SNR)	Interpretability	Suitability for Mean-Reversion
WaveNet	Short–mid range, FIR via dilated convolution stack	Fixed-scale gated filters	High (parallel convolutions)	Low latency ($O(1)$ per step)	High (deterministic filters)	Medium	Moderate. FIR structure with transparent dilation schedule.	Stable multi-scale FIR; effective for short cycles
DPRNN	Mid–long range, hybrid (FIR intra-chunk, IIR inter-chunk)	Chunk-wise recurrent state	Moderate (sequential blocks)	Moderate latency	Moderate (recurrent structure)	Medium–High	Moderate. Local recurrence gives partial interpretability.	Balances local/global context; suitable for mid-horizon dynamics
Informer	Long range, non-recursive FIR via sparse attention	Content-adaptive attention over inputs	High (batched attention)	High latency (non-streamable)	Low–Moderate (outlier sensitivity)	Low	Low. Sparse attention paths are opaque in depth.	Captures global triggers; fragile under low SNR or latency constraints
DKF	Infinite memory, IIR via stochastic latent recursion	Amortized variational posterior $q_\phi(x_t \mid y_{\leq t})$	Low–Moderate (expensive to train due to RNN structure)	Moderate latency	Moderate (posterior stability)	High	Moderate. Latent state interpretable under structured priors.	Handles noisy dynamics with calibrated uncertainty
S4	Infinite memory, IIR via stable state-space recurrence	Fixed or spectrally tunable via pole-residue structure	High (FFT or scan)	Low latency ($O(1)$ scan or FFT conv)	High (provably stable kernel)	High	High. Poles/residues provide spectral and temporal insight.	Stable long-memory filtering; well-suited for low-SNR signal tracking

5 Mean Reversion Trading with Filters

We first describe a minimal threshold-based trading strategy, as a stochastic control rule that is extensible to systems such as Leung and Li (2015) and Lipton and Prado (2020). Each model's filtered output $\hat{y}_t$ is converted into portfolio positions by reacting to deviations between the filtered fair value and the observed market price. Model performance is then evaluated along two dimensions: 1. signal stability, quantifying the smoothness and turnover of portfolio allocations, and 2. trading performance, captured through examining the Sharpe ratio, excess returns, and hit rate from trading with each model. In this way, the spread or relative mispricing is connected to trading performance, providing a pathway for model optimisation and calibration.

We compare exponential moving average filters and 2-layer CNN models against Wavenet, as a complex CNN model with more effective context representation and utilisation. Meanwhile, we implement a DKF model as a method of state space and Bayesian estimation. The models are applied without systematic parameter calibration. In order to perform systematic calibration, one may build a control system connecting the model and trading parameters with trading performance metrics and defining optimality conditions. This paper emphasises the methodological aspects of mean reversion trading with signal filters. The optimisation and out-of-sample evaluation/comparison of the models should be addressed in a separate paper.

5.1 Trading on price spreads

Define a state-dependent control rule $f : (\hat{y}_t, y_t) \to w_t$ acting on the spread between the filtered $\hat{y}_t$ and the spot price y_t to a portfolio position $w_t \in [-1, 1]$. The relative mispricing δ_t is:

$$\delta_t = \frac{y_t - \hat{y}_t}{\hat{y}_t}.$$

Let $\tau_{\text{entry}} > 0$ be the entry threshold and $\alpha > 0$ be a scaling parameter. The allocation rule enters a long (respectively, short) position when the mispricing magnitude exceeds τ_{entry}, scales the position continuously up to a maximum of 1.

$$w_t = \begin{cases} -\operatorname{sgn}(\delta_t) \min\{1, \alpha \max(0, |\delta_t| - \tau_{\text{entry}})\}, & |\delta_t| \geq \tau_{\text{entry}} \\ 0 & |\delta_t| < \tau_{\text{entry}} \end{cases}.$$

5.2 Evaluation metrics

Signal stability and turnover The asset allocation should be resistant to small changes and fluctuations. We measure stability via the average absolute change in position, i.e. the turnover,

$$\text{Turnover} = \frac{1}{T-1} \sum_{t=2}^{T} |w_t - w_{t-1}|, \tag{5.1}$$

where lower values indicate smoother and more stable signals. Equivalently, its complement $1 -$ Turnover can be interpreted as a simple signal stability index.

Transaction-cost-adjusted strategy returns Let $r_t^{\text{spot}} = \frac{y_t - y_{t-1}}{y_{t-1}}$ denote the spot return over $[t-1, t]$, and define the (gross) strategy return for model m as

$$r_t^{(m)} = w_{t-1} r_t^{\text{spot}}. \tag{5.2}$$

To incorporate trading frictions, we assume proportional transaction costs that are linear in position changes. Let $c > 0$ denote the cost per unit of turnover, and define the per-period cost as

$$\text{TC}_t = c|w_t - w_{t-1}|. \tag{5.3}$$

The transaction-cost-adjusted (net) strategy return is then

$$r_{t,\text{net}}^{(m)} = r_t^{(m)} - \text{TC}_t = w_{t-1} r_t^{\text{spot}} - c|w_t - w_{t-1}|. \tag{5.4}$$

All performance measures below are computed using the net return series $\{r_{t,\text{net}}^{(m)}\}$.

Hit rate during active positions Since the strategy only takes risk when $|w_t| > 0$, it is informative to measure how often these active positions are profitable. Define the indicator of a positive net return as

$$\mathbb{I}_t^{(m)} = \mathbb{I}\{r_{t,\text{net}}^{(m)} > 0\},$$

and the indicator of an active position as

$$H_t = \mathbb{I}\{|w_{t-1}| > 0\}.$$

The hit rate of model m is defined as the fraction of periods with a positive net return, conditional on holding a non-zero position,

$$\text{HitRate}^{(m)} = \frac{\sum_{t=1}^{T} H_t \mathbb{I}_t^{(m)}}{\sum_{t=1}^{T} H_t}. \tag{5.5}$$

Excess return, Sharpe ratio, and Sortino ratio Let r_f denote the risk-free rate per period.[1] The average excess return of model m is

$$\text{ER}^{(m)} = \frac{1}{T} \sum_{t=1}^{T} (r_{t,\text{net}}^{(m)} - r_f). \tag{5.6}$$

Let $\bar{r}_{\text{net}}^{(m)}$ and $\sigma(r_{t,\text{net}}^{(m)})$ denote the sample mean and standard deviation of the net returns. If N

[1] In our intraday setting, we set $r_f = 0$, assuming the risk-free return over each interval is negligible.

denotes the number of return periods in a year, the annualised Sharpe ratio is

$$\mathrm{SR}_{\mathrm{ann}}^{(m)} = \sqrt{N}\frac{\bar{r}_{\mathrm{net}}^{(m)} - r_f}{\sigma(r_{t,\mathrm{net}}^{(m)})}. \tag{5.7}$$

To distinguish between upside and downside risk, we also report the Sortino ratio. Define the downside deviation as

$$\sigma_-(r_{t,\mathrm{net}}^{(m)}) = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(\min\{r_{t,\mathrm{net}}^{(m)} - r_f, 0\})^2}, \tag{5.8}$$

and the annualised Sortino ratio as

$$\mathrm{SoR}_{\mathrm{ann}}^{(m)} = \sqrt{N}\frac{\bar{r}_{\mathrm{net}}^{(m)} - r_f}{\sigma_-(r_{t,\mathrm{net}}^{(m)})}. \tag{5.9}$$

5.3 Results

Models are fitted on 2-minute intraday data of the S&P 500 index. All neural network filters are trained on context windows of length 512. We constrain them to the same LAFO parameters K, λ, and ℓ_2 penalty. The resulting models filter at similar frequencies. The relative mispricing signal, δ, for each model is therefore similar in magnitude, and we may compare different models sharing τ_{entry} and α on the same dataset/asset. Figure 3 compares the filtered signals, confirming that the fitted models filter data at similar frequencies with some local differences. The jump after index 200 is caused by an overnight price change.

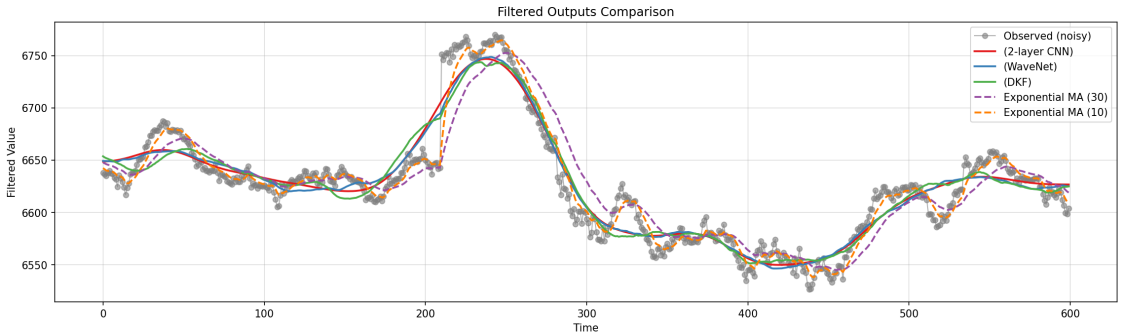


Figure 3: Raw filter signal comparison.

Figure 4 plots the signals and trading portfolio of the Wavenet model on the last 600 samples of the dataset, the same range of data as in Figure 3. The model generates consistent profit after accounting for proportional trading costs. This figure shows the transformations from the relative mispricing signal to portfolio allocations and trading performance. This paper offers insight to how model and optimisation objective selection influence the filtered signals, but more effort is required to understand how to optimise the relative mispricing signal for maximum trading returns. A first step may be investigating the optimal rate of reversion or multiple trading (entry, profit taking,

stop-loss) thresholds for δ.

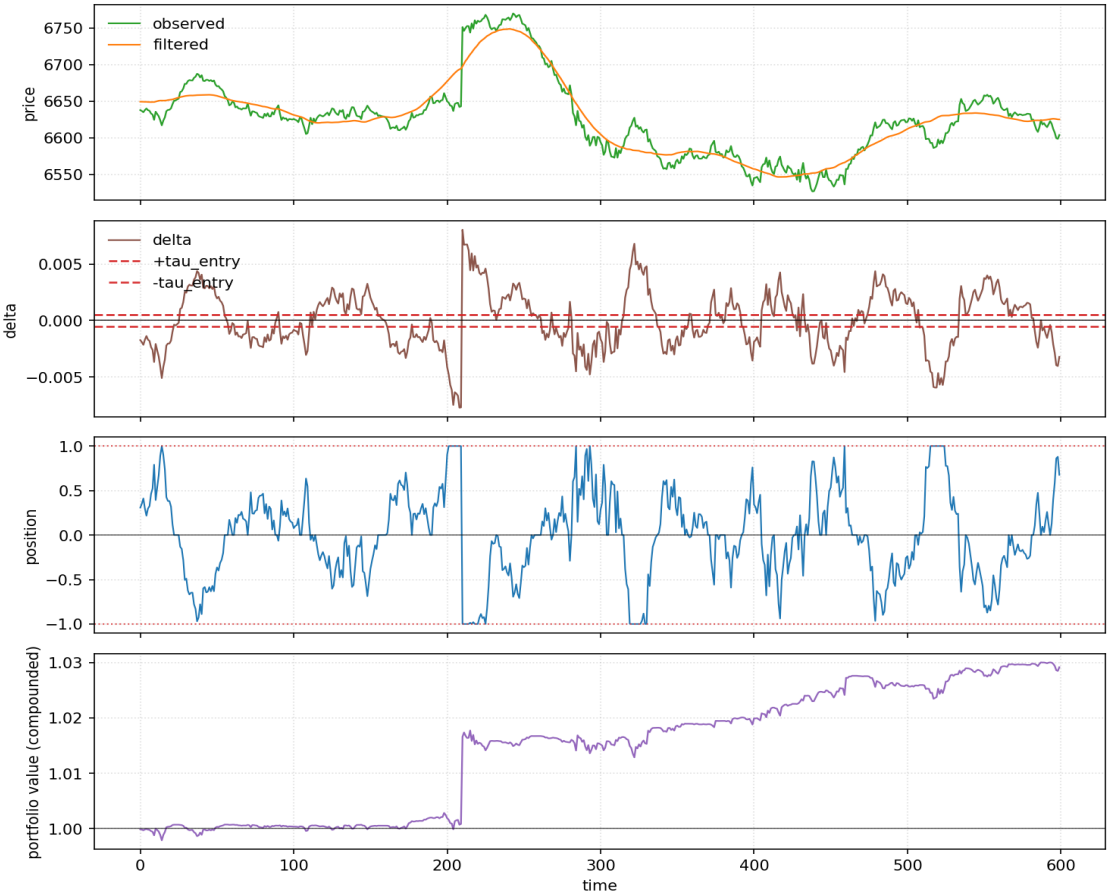


Figure 4: Wavenet portfolio, S&P 500, 2-min frequency, trading on last 600 samples

The trading metrics table 3 reveals differences in trading performance, even when models filter at similar frequencies. Neural network models outperform the exponential moving average models. Observations suggest that model complexity does not improve trading results for this case study. The 2-layer CNN model is smaller, more stable, and interpretable. It is expected to work well on smaller datasets (about 6000 samples for this experiment). In contrast, the complexity of DKF and Wavenet models suggests that they are much more data-intensive. A receptive field length of 512, as we use in this case study, may be large for the typical kernel size of simple CNN models, but advanced models are expected to distinguish themselves on larger datasets and longer contexts.

Table 3: Performance, stability, and shared-configuration metrics. Mean reversion trading on spot-filter price spread.

S&P 500, 2-min frequency, 2 months

Model	Turnover	Sharpe Ratio	Excess Return	Hit Rate	Sortino Ratio
EMA(10)	0.037	−18.265	−0.06370	0.317	−25.099
EMA(30)	0.055	−14.761	−0.09431	0.370	−19.591
2-layer CNN	0.050	11.049	0.10530	0.479	39.134
Wavenet	0.052	8.005	0.07041	0.436	24.167
DKF	0.052	4.920	0.04153	0.432	12.890

Shared trading-rule/model configuration

LAFO window K	λ (smoothing penalty)	τ_{entry}	α	trading cost c
80	20.0	0.0005	200.0	0.0003(3bp)

A robust, out-of-sample, statistically rigorous systematic study of the models on large and diverse datasets would help rationalise the usage of advanced filtering models. In this paper, however, our focus is primarily methodological: we emphasise how model calibration and evaluation should be tied to trading performance metrics. We show a mathematically coherent framework or pathway connecting filtering models to their trading performances. Each stage of the framework can be formalised and analysed mathematically.

6 Conclusion and Future Works

This paper shows a framework for mean reversion trading though fair value filtering, using advanced neural network models trained on trading-oriented objective functions. We propose, with theoretical and empirical support, penalised LAFO trading objectives to train network models for stable, regime-aware, adaptive low-pass filtering. We evaluate advanced neural networks that implement different context retrieval mechanisms. Advanced architectures may improve inference reliability and investment performance, but this paper cannot conclusively demonstrate this. In fact, for the empirical experiments in this paper, interpretability and inference stability may be the reason the simple CNN model has achieved better performance compared to the DKF and Wavenet models, which require more careful tuning and adjustments.

We define a mean reversion trading rule that is extensible to a full stochastic control problem reacting to the relative mispricing or spread between spot and filtered signals. The trading system provides a pathway connecting models and their filtered signals to trading performances. This framework is therefore extensible to support model calibration and optimisation around trading-oriented metrics.

Directions of improvement:

- By systematically investigating the interaction between the filter signal and trading outcome, we may derive optimal entry, exit, or threshold trading levels. One may also examine the optimal trading horizon or frequency for a specific low-pass filter weighting. These analyses may be possible by connecting the statistical properties of the spread between filtered and spot prices (reversion rate, stationarity) to trading metrics such as mean returns and variances through mathematical formalities.

- Filtering models should be compared out-of-sample, retrained over time, and evaluated on multiple datasets. Significance testing can help compare different models. This answers whether advanced, modern neural network architectures improve trading performances.

- While long-context retrieval mechanisms allow neural networks to infer nonlinear changes effectively, models in this paper do not consist specialised infrastructure to detect change points. The models may improve by conditioning on explicit change point statistics, other regime changing models, or one may embed a specialised change point detection structure within the models.

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